

習題 2.9

$$y'(t) = \int_0^t f(\xi) \varphi'(t-\xi) d\xi + f(t) \varphi(0) - \cancel{\int_0^t f(\xi) \varphi(t-\xi) d\xi}$$

$$\begin{aligned} y' + \lambda y &= f(t) + \int_0^t f(\xi) \varphi'(t-\xi) d\xi + \lambda \int_0^t f(\xi) \varphi(t-\xi) d\xi \\ &= f(t) + \int_0^t f(\xi) (\varphi'(t-\xi) + \lambda \varphi(t-\xi)) d\xi = f(t) \end{aligned}$$

習題 2.10

設 $-u_2'' + \lambda u_2 = 0$, $u_2(0) = 0$, $u_2'(0) = b$, 則 $u_2(t) = b \varphi(t) = \cancel{M_6(t)} \cancel{M_6(t)} M_6(t)$

設 則 $u_1(t) = \frac{dM_4(t)}{dt} = a \varphi'(t)$ $[-u_1'' + \lambda u_1 = 0, u_1(0) = a, u_1'(0) = 0]$

$$u_3(t) = -\int_0^t M_{52}(t-\tau) d\tau = -\int_0^t f(\tau) \varphi(t-\tau) d\tau \quad [-u_3'' + \lambda u_3 = f(t), u_3(0) = 0, u_3'(0) = 0]$$

$$\Rightarrow y(t) = a \varphi'(t) + b \varphi(t) - \int_0^t f(\tau) \varphi(t-\tau) d\tau$$

習題 2.11

$$\text{設 } \omega = \frac{1}{m}, \quad v = mu.$$

$$f(x,t) = \partial_t(mu) - \partial_{xx}(mu) = m \partial_t u + u \partial_t m - \partial_x(m \partial_x u + u \partial_x m)$$

$$= m \partial_t u + u \partial_t m - m \partial_{xx} u - \partial_x u \partial_x m - u \partial_{xx} m - \partial_x m \partial_x u$$

$$= m(\partial_t u - \partial_{xx} u - \frac{2 \partial_x m}{m} \partial_x u + \frac{\partial_{xx} m}{m} u) + \cancel{u \partial_t m} \frac{\partial_t m - \partial_{xx} m}{m} u$$

$$\begin{cases} -2 \frac{\partial_x m}{m} = a \\ \frac{\partial_t m - \partial_{xx} m}{m} = b \end{cases}$$

$$\Rightarrow \begin{cases} \partial_x m = -\frac{1}{2} m a \\ \frac{\partial_t m - \partial_x(\partial_x m)}{m} = \frac{\partial_t m + \frac{a}{2} \partial_x m}{m} = \frac{\partial_t m - \frac{a^2}{4} m}{m} = b \end{cases}$$

$$\Rightarrow \begin{cases} \partial_x m = -\frac{1}{2} m a \\ \partial_t m = \frac{a^2}{4} m + b m \end{cases} \Rightarrow \begin{cases} m = C e^{-\frac{a}{2} x} + \cancel{\frac{a^2}{4} \varphi(t)} \\ \varphi'(t) = \cancel{\frac{a^2}{4} (C e^{-\frac{a}{2} x} + 2 C e^{-\frac{a}{2} x} \varphi(t) + \varphi^2(t))} + b (C e^{-\frac{a}{2} x} + b \varphi(t)) \end{cases}$$

可得 $C=0$

$$\begin{cases} m = \varphi(t) e^{-\frac{a}{2} x} \\ \varphi'(t) e^{-\frac{a}{2} x} = \varphi(t) e^{-\frac{a}{2} x} \left(\frac{a^2}{4} + b \right) \end{cases}$$

$$\Rightarrow m = C e^{-\frac{a}{2} x + (\frac{a^2}{4} + b)t}$$

$$\Rightarrow \omega(x,t) = \frac{1}{m(x,t)} = C^{-1} e^{\frac{a}{2} x - (\frac{a^2}{4} + b)t}$$

$$\text{取 } C=1, \text{ 可得 } \omega = e^{\frac{a}{2} x - (\frac{a^2}{4} + b)t}$$

習題 2.12

設 $\varphi(x, y, z) = f(x) + g(y)$

$$u(x, y, z, t) = \frac{1}{4\pi a^2 t} \iint_{|\vec{y}-\vec{x}|=at} \varphi(x, y, z) dS = \frac{1}{4\pi a^2 t} \iint_{\sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2} = at} [f(\xi) + g(\eta)] dS$$

$$u(x, y, z, t) = \partial_t \left[\frac{1}{4\pi a^2 t} \iint_{|\vec{y}-\vec{x}|=at} (f(\xi) + g(\eta)) dS \right] + \frac{1}{4\pi a^2 t} \iint_{|\vec{y}-\vec{x}|=at} (\psi(\eta) + \psi(\zeta)) dS$$

習題 2.13

$$u(x, y, t) = \frac{1}{2\pi a} \left[\partial_t \iint_{\Sigma at(x, y)} t \sin y d\sigma \right]$$

$$u(x, y, t) = \frac{1}{2\pi a} \left[\partial_t \iint_{\Sigma at(x, y)} \frac{\xi_x^2 + \sin \xi_y d\xi d\eta}{\sqrt{a^2 t^2 - [(x-\xi)^2 + (y-\eta)^2]}} + \iint_{\Sigma at(x, y)} \frac{\sin \xi_y d\xi d\eta}{\sqrt{a^2 t^2 - [(x-\xi)^2 + (y-\eta)^2]}} \right. \\ \left. + \int_0^t \iint_{\Sigma a(t-\tau)(x, y)} \frac{\tau \sin \xi_y d\xi d\eta}{\sqrt{a^2 (t-\tau)^2 - [(x-\xi)^2 + (y-\eta)^2]}} d\tau \right]$$

習題 2.12

把 $u(x, y, z, t)$ 分解成 $m(x, y, z, t) + n(x, y, z, t) + p(x, y, z, t)$

其中 $\begin{cases} \partial_{tt} m - a^2 \partial_{xx} m = 0 \\ m(x, y, z, t) = \tilde{m} \end{cases} \quad \tilde{m}(x, t) = m(x, y, z, t) \quad \begin{cases} \partial_{tt} \tilde{m} - a^2 \partial_{xx} \tilde{m} = 0 \\ \tilde{m}(x, 0) = f(x) \\ \partial_t \tilde{m}(x, 0) = 0 \end{cases}$

$\tilde{n}(y, t) = n(x, y, z, t) \quad \begin{cases} \partial_{tt} \tilde{n} - a^2 \partial_{yy} \tilde{n} = 0 \\ \tilde{n}(y, 0) = g(y) \\ \partial_t \tilde{n}(y, 0) = \psi(y) \end{cases}$

$\tilde{p}(z, t) = p(x, y, z, t) \quad \begin{cases} \partial_{tt} \tilde{p} - a^2 \partial_{zz} \tilde{p} = 0 \\ \tilde{p}(z, 0) = 0 \\ \partial_t \tilde{p}(z, 0) = \psi(z) \end{cases}$

$$\tilde{p}(z, t) = \frac{1}{2a} \int_{z-at}^{z+at} \psi(\xi) d\xi$$

$$\tilde{n}(y, t) = \frac{1}{2a} [g(y+at) + g(y-at)] + \frac{1}{2a} \int_{y-at}^{y+at} \psi(\xi) d\xi$$

$$\tilde{m}(x, t) = \frac{1}{2} [f(x+at) + f(x-at)]$$

$$u(x, y, z, t) = \frac{1}{2} (f(x+at) + g(y+at) + f(x-at) + g(y-at)) + \frac{1}{2a} \left[\int_{y-at}^{y+at} \psi(\xi) d\xi + \int_{z-at}^{z+at} \psi(\xi) d\xi \right]$$

習題 2.13

設 $u(x, y, t)$ 分解成 $m(x, y, t) + n(x, y, t)$

$$\text{其中 } \tilde{m}(x, t) = m(x, y, t) \quad \begin{cases} \partial_{tt} \tilde{m} - a^2 \partial_{xx} \tilde{m} = 0 \\ \tilde{m}(x, 0) = x^2 \\ \partial_t \tilde{m}(x, 0) = 0 \end{cases} \Rightarrow \tilde{m}(x, t) = \frac{1}{2}((x+at)^2 + (x-at)^2) = x^2 + a^2 t^2$$

$$\tilde{n}(y, t) = n(x, y, t) \quad \begin{cases} \partial_{tt} \tilde{n} - a^2 \partial_{yy} \tilde{n} = \psi \sin y \\ \tilde{n}(y, 0) = 0 \\ \partial_t \tilde{n}(y, 0) = \sin y \end{cases}$$

$$\begin{aligned} \Rightarrow \tilde{n}(y, t) &= \frac{1}{2a} \left[\int_{y-at}^{y+at} \sin \xi d\xi + \int_0^t \int_{y-a(t-\tau)}^{y+a(t-\tau)} \tau \sin \xi d\xi d\tau \right] \\ &= \frac{1}{2a} \left[\cos(y-at) - \cos(y+at) + \int_0^t \tau [\cos(y-a(t-\tau)) - \cos(y+a(t-\tau))] d\tau \right] \\ &= \frac{1}{2a} \left[2\sin y \sin at + \frac{1}{a} \int_0^t \tau [\cos(at-\tau+y) - \cos(at-\tau-y)] d(at) \right] \\ &= \frac{\sin y \sin at}{a} + \frac{1}{2a^2} \left[\frac{2t \sin y}{2} - \int_0^t [\sin(at-\tau+y) - \sin(at-\tau-y)] d\tau \right] \\ &= \frac{\sin y \sin at}{a} + \frac{1}{2a^2} [\cos(at-\tau+y) - \cos(at-\tau-y)]_0^t + \frac{t \sin y}{a^2} \\ &= \frac{\sin y \sin at}{a} + \frac{1}{2a^2} [-\cos(y-at) + \cos(y+at)] + \frac{t \sin y}{a^2} \\ &= \frac{\sin y \sin at}{a} - \frac{1}{a^2} (\sin y \sin at) + \frac{t \sin y}{a^2} \end{aligned}$$

$$\Rightarrow u(x, y, z, t) = x^2 + a^2 t^2$$

$$\Rightarrow u(x, y, z, t) = x^2 + a^2 t^2 + \frac{a^2 - 1}{a^3} \sin y \sin at + \frac{t \sin y}{a^2}$$

習題 2.14

$$u|_{x-at=0} = \varphi(x) \Rightarrow u(at, t) = \varphi(at)$$

$$\text{記 } \xi = x+at, \eta = x-at$$

$$\partial_t u = (\partial_\xi u) a + \partial_\eta u \cdot (-a), \quad \partial_x u = \partial_\xi u + \partial_\eta u$$

$$\partial_{tt} u = a^2 (\partial_{\xi\xi} u + \partial_{\eta\eta} u), \quad \partial_{xx} u = \partial_{\xi\xi} u + 2\partial_{\xi\eta} u + \partial_{\eta\eta} u$$

$$= a(a\partial_{\xi\xi} u - a\partial_{\eta\eta} u + 2\partial_{\xi\eta} u - a\partial_{\xi\eta} u + a\partial_{\eta\eta} u)$$

$$\Rightarrow \partial_{\xi\eta} u = 0 \Rightarrow u(x, t) = F(x+at) + G(x-at)$$

$$\Rightarrow u|_{x-at=0} = F(2x) + G(0) = \varphi(x)$$

$$\Rightarrow u|_{x+at=0} = F(0) + G(2x) = \psi(x)$$

$$\Rightarrow u(x, t) = \varphi\left(\frac{x+at}{2}\right) + \psi\left(\frac{x-at}{2}\right) - F(0) - G(0)$$

$$= \varphi\left(\frac{x+at}{2}\right) + \psi\left(\frac{x-at}{2}\right) - \varphi(0)$$

$$\# \varphi: -1 \leq \frac{x-at}{2} \leq 0 \leq \frac{x+at}{2} \leq 2$$

$$-2 \leq x-at \leq 0 \leq x+at \leq 4$$

$$\Leftrightarrow \max\{-2+at, at\} \leq x \leq \min\{4-at, at\}$$

習題 2.15

記 $\vec{y} = (y_1, y_2)$, $\vec{x} = (x_1, x_2)$, $d\vec{y} = dy_1 dy_2$,

$\varphi(\vec{y}) = \hat{\varphi}(y_1)$, $\varphi(\vec{y}) = \hat{\varphi}(y_1)$

$$u(x_1, t) = \frac{1}{2\pi a} \left[\partial_t \iint_{|\vec{y}-\vec{x}|=\sqrt{a^2 t^2 - |\vec{y}-\vec{x}|^2}} \frac{\hat{\varphi}(y_1) dy_1 dy_2}{\sqrt{a^2 t^2 - |\vec{y}-\vec{x}|^2}} + \iint_{|\vec{y}-\vec{x}| \leq at} \frac{\hat{\varphi}(y_1) dy_1 dy_2}{\sqrt{a^2 t^2 - |\vec{y}-\vec{x}|^2}} \right]$$

記 $y_1 - x_1 = at y_1 \Leftrightarrow y_1 = x_1 + at y_1$ ($|y_1| \leq 1$)

$y_2 - x_2 = at y_2 \Rightarrow y_2 = x_2 + at y_2$ ($|y_2| \leq 1$) 且 $y_1^2 + y_2^2 \leq 1$

$$\begin{aligned} \Rightarrow u(x_1, t) &= \frac{1}{2\pi a} \left[\iint_{|(y_1, y_2)| \leq 1} \frac{\hat{\varphi}(x_1 + at y_1) a^2 t^2 dy_1 dy_2}{at \sqrt{1 - y_1^2 - y_2^2}} + \iint_{|(y_1, y_2)| \leq 1} \frac{\hat{\varphi}(y_1) a^2 t^2 dy_1 dy_2}{at \sqrt{1 - y_1^2 - y_2^2}} \right] \\ &= \frac{1}{2\pi} \left[\partial_t \int_{-1}^1 \hat{\varphi}(x_1 + at y_1) \cdot t \int_{-\sqrt{1-y_1^2}}^{\sqrt{1-y_1^2}} \frac{dy_2}{\sqrt{(1-y_1^2)-y_2^2}} dy_1 \right. \\ &\quad \left. + \int_{-1}^1 \hat{\varphi}(x_1 + at y_1) \cdot t \int_{-\sqrt{1-y_1^2}}^{\sqrt{1-y_1^2}} \frac{dy_2}{\sqrt{(1-y_1^2)-y_2^2}} dy_1 \right] \\ &= \frac{1}{2\pi} \left[\partial_t \int_{-1}^1 \hat{\varphi}(x_1 + at y_1) t \arcsin \left(\frac{y_2}{\sqrt{1-y_1^2}} \right) \Big|_{-\sqrt{1-y_1^2}}^{\sqrt{1-y_1^2}} dy_1 + \int_{-1}^1 \hat{\varphi}(x_1 + at y_1) t \arcsin \left(\frac{y_2}{\sqrt{1-y_1^2}} \right) \Big|_{-\sqrt{1-y_1^2}}^{\sqrt{1-y_1^2}} dy_1 \right] \\ &= \frac{1}{2} \left[\partial_t \int_{-1}^1 \hat{\varphi}(x_1 + at y_1) t dy_1 + \int_{-1}^1 \hat{\varphi}(x_1 + at y_1) t dy_1 \right] \\ &= \frac{1}{2} \left[\partial_t \int_{x_1-at}^{x_1+at} \varphi(y_1) \cdot \frac{1}{a} dy_1 + \int_{x_1-at}^{x_1+at} \hat{\varphi}(y_1) \cdot \frac{1}{a} dy_1 \right] \\ &= \frac{1}{2a} \left[\varphi(x_1 + at) + \varphi(x_1 - at) \right] + \frac{1}{2a} \int_{x_1-at}^{x_1+at} \hat{\varphi}(y_1) dy_1 \end{aligned}$$

$$\Rightarrow u(x, t) = \frac{1}{2} (\varphi(x+at) + \varphi(x-at)) + \frac{1}{2a} \int_{x-at}^{x+at} \hat{\varphi}(\xi) d\xi$$

習題 2.16

(1) 取 $w(x, t) = xg(t)$, $u = v + w$

$$\partial_x v(0, t) = \partial_x u(0, t) - \partial_x w(0, t) = g(t) - g(t) = 0$$

$$\partial_t v(x, 0) = \partial_t u(x, 0) - \partial_t w(x, 0) = \psi(x) - xg'(0) = \psi_1(x)$$

$$v(x, 0) = u(x, 0) - w(x, 0) = \varphi(x) - xg(0) = \varphi_1(x)$$

$$\partial_{tt} v - a^2 \partial_{xx} v = \partial_{tt} u - a^2 \partial_{xx} u - \partial_{tt} w + a^2 \partial_{xx} w = f(x, t) - xg''(t) = f_1(x, t)$$

(2) 對 f_1, φ_1, ψ_1 作偶延拓.

$$\bar{f}_1(x, t) = \begin{cases} f_1(x, t), & x \geq 0 \\ f_1(-x, t), & x < 0 \end{cases} \quad \bar{\varphi}_1(x, t) = \begin{cases} \varphi_1(x, t), & x \geq 0 \\ \varphi_1(-x, t), & x < 0 \end{cases} \quad \bar{\psi}_1(x, t) = \begin{cases} \psi_1(x, t), & x \geq 0 \\ \psi_1(-x, t), & x < 0 \end{cases}$$

$$\begin{cases} \partial_{tt} v - a^2 \partial_{xx} v = \bar{f}_1(x, t), & x \in \mathbb{R}, t > 0 \\ v(x, 0) = \bar{\varphi}_1(x, t) \\ \partial_x v(x, 0) = \bar{\psi}_1(x, t) \end{cases}$$

因為 f_1, φ_1, ψ_1 都是關於 x 的偶函數, 所以 v 也是關於 x 的偶函數

$$\Rightarrow \partial_x v(0, t) = 0.$$

(3)

$$v(x, t) = \frac{1}{2} (\bar{\varphi}_1(x+at) + \bar{\varphi}_1(x-at)) + \frac{1}{2a} \int_{x-at}^{x+at} \bar{\psi}_1(\xi) d\xi + \frac{1}{2a} \int_0^t \int_{x-a(t-\tau)}^{x+a(t-\tau)} \bar{f}_1(\xi, \tau) d\xi d\tau$$

當 $x \geq at$ 時, $v(x, t) = \frac{1}{2} (\varphi_1(x+at) + \varphi_1(x-at)) + \frac{1}{2a} \int_{x-at}^{x+at} \psi_1(\xi) d\xi + \frac{1}{2a} \int_0^t \int_{x-a(t-\tau)}^{x+a(t-\tau)} f_1(\xi, \tau) d\xi d\tau$

當 $x < at$ 時, $v(x, t) = \frac{1}{2} (\varphi_1(at+x) + \varphi_1(at-x)) + \frac{1}{2a} \int_{at-x}^{at+x} \psi_1(\xi) d\xi$

$$+ \frac{1}{2a} \left[\int_0^{t-\frac{x}{a}} \int_{a(t-\tau)-x}^{a(t-\tau)+x} f_1(\xi, \tau) d\xi d\tau + \int_{t-\frac{x}{a}}^t \int_{x-a(t-\tau)}^{x+a(t-\tau)} f_1(\xi, \tau) d\xi d\tau \right]$$

$$+ \frac{1}{2a} \left[\int_0^{t-\frac{x}{a}} \left(\int_0^{a(t-\tau)-x} f_1(\xi, \tau) d\xi + \int_0^{a(t-\tau)+x} f_1(\xi, \tau) d\xi \right) d\tau \right].$$

(4) $\begin{cases} \partial_x u(0, 0) = g(0) \\ \partial_x u(a, 0) = \varphi'(0) \end{cases} \Rightarrow g(0) = \varphi'(0) = \partial_x u(0, 0)$

$$\begin{cases} \partial_{tx} u(0, 0) = g'(0) \\ \partial_{xt} u(0, 0) = \psi'(0) \end{cases} \Rightarrow g'(0) = \psi'(0) = \partial_{tx} u(0, 0) = \partial_{xt} u(0, 0)$$

$$\begin{aligned} \partial_x f(0, 0) &= \partial_{xtt} u(0, 0) - \partial_{xxx} u(0, 0) \\ &= g''(0) - a^2 \psi'''(0) \end{aligned}$$